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Department of Pharmacognosy
School of Pharmacy

Laboratory Manual

Pharmacognosy I (Phar 224)

10-10
12/12/14
10/11/15

EXPERIMENT I

EXTRACTION

Introduction: Crude drugs are vegetable or animal drugs that have undergone only the process of collection and drying. As a result, crude drugs are not usually directly used as therapeutic agents; more often their active principles are separated by various means and are employed in a more purified form. The process of drug extraction is a generally accepted method of obtaining these active principles. Extraction removes only those substances that can be dissolved in the solvent used for the extraction. The undissolved portion of the drug that remains after the extraction process is completed is called the marc. The extract will, usually, contain a complex mixture and is thus itself needs further purification.

There exist many techniques of extraction. These include refluxing, maceration, percolation and soxhlet continuous extraction. The particular method of extraction chosen is dependent on factors like the type of plant material to be extracted, the purpose of extraction, physicochemical properties of the extract, availability of solvent and apparatus, etc. In this experiment, refluxing, maceration, percolation and soxhlet extraction are employed in the extraction of different constituents from the leaves of *Calpurnea aurea*.

Apparatus and Materials: Plant material, reflux apparatus, Soxhlet apparatus, solvent, maceration chamber, filtering apparatus, heating stove, porcelain dishes.

Procedure:

Maceration:

1. Powder 30 g of *Calpurnia* leaves using mortar and pestle and transfer this to a percolator whose bottom has been plugged with cotton.
2. Add enough amount of 70% alcohol into the maceration chamber and leave for 72 hrs with occasional stirring.
3. Collect the percolate in flask and wash the marc with about 100 ml were solvent
4. Reduce the volume of the percolate under reduced pressure
5. Transfer the remaining extract to a weighed porcelain dish and evaporate to dryness over a water bath.
6. Calculate the percentage yield. Reserve the extract for future experiments

Refluxing

1. Powder 20 g of the plant material using a mortar & pestle.
2. Transfer the powder in to a 250 ml-flask and add about 100 ml of 70% methanol.
3. Attach a reflux condenser and reflux for 48 hrs.
4. Cool the contents of the flask under running tap water and filter
5. Concentrate the filtrate under reduced pressure and transfer the concentrate to a weighed porcelain dish and evaporate to dryness over a water bath.
6. Calculate the percentage yield, Reserve the residue for future experiments.

Soxhlet Extraction

1. Powder 20 g of the plant material using a mortar & pestle
2. Transfer the powder in to the thimble provided and cover the top with cotton.
3. Place the thimble in the extraction chamber of the Soxhlet. Add 200 ml of alcohol into a 250 ml flask and heat for about 2 hours.
4. Remove the flask and cool under running water and concentrate the extract under reduced pressure and transfer it to a weighed porcelain dish and evaporate to dryness over a water bath.
5. Calculate the percentage yield and reserve the residue the future experiments.

EXPERIMENT II

COLUMN CHROMATOGRAPHY

Introduction: Chromatography is physical method of separation process based on the variation in the rate at which different components of a mixture migrate through a stationary phase under the influence of a mobile phase. During the chromatographic process, solute molecules move back and forth between two phases. The mobile phase carries them forward but they remain stationary during the time they spend in the stationary phase. Rate of migration is thus determined by the solutes' distribution ratio or in other words the proportion of time they spend in the stationary phase. A process by which a solute is transferred from a mobile phase to a stationary phase is called sorption. And chromatographic techniques are based on four mechanisms of sorption namely adsorption, partition, ion exchange, and exclusion.

In this experiment a glass column is packed with silica gel and then the sample is applied at the top. Separation of components is brought about as the solvent percolates downward carrying with it components of the sample. Also a variety of factors that enhance the efficiency of separation are considered.

Apparatus and Materials: glass column, silica gel, plant extract, sand, filter paper, beaker, solvent, solvent reservoir and Erlenmeyer flasks.

Procedure:

1. Preparation of the column

1. Set the column with sintered glass at the bottom in a vertical position by clamping it to a stand.
2. Suspend the adsorbent in the solvent that is to be used as an eluent and carefully add the suspension in to the column gently a little at a time.
3. Allow the adsorbent to settle and open the stopcock to let the excess solvent flow out until it is only a few centimeters above the column of packing material.
4. Repeat the packing process until the adsorbent layer has reached a reasonable height.
5. Place a filter paper on top of the adsorbent. You should make sure that the solvent layer is still 3-4 cm above the top of the adsorbent.
6. Place a sand layer on top of the filter paper covering the top of the adsorbent. Now the solvent layer should be 1.5-2.0 cm above the sand layer.

II. Preparation and application of the sample solution

1. Redissolve the extract you obtained in experiment I in a small quantity of 70% ethanol to form a slightly viscous solution.
2. Pour the extract carefully portion-wise in to the column set above.
3. Fill reservoir with a stopcock on top of the column.
4. Add sufficient amount of ethanol into the reservoir and open its stopcock.
5. When the remaining portion of the column is filled, start to regulate the flow of the solvent by manipulating the stopcock at the bottom of the column.
6. Go on to collect different fractions in different receivers while recording your observations as the process of uniform percolation proceeds.

EXPERIMENT III

THIN LAYER CHROMATOGRAPHY (TLC)

Introduction: This chromatographic technique was devised after column and paper chromatography. It combines the better resolution power of column chromatography with ease of manipulation of paper. It is a technique of separation and identification of chemical substances by a moving solvent on a thin layer of absorbent. The mechanisms of resolution are similar qualitatively to those involved in paper chromatography. However, quantitatively their relative magnitude differs greatly. This particular technique is preferable in the analysis of lipids, alkaloids, flavonoids, etc. The ease with which one can change the type of absorbent and use various methods of detection have made this technique popular.

In this experiment spots of a plant extract are applied on the TLC plate and developed. The resolved components are then studied under visible & UV lights and after spraying with reagents. R_f values of the different spots are calculated and a tracing of the developed plate is prepared.

Apparatus and Materials: TLC plate, silica gel, solvent system, chromatographic tank, capillary tubes, beaker, UV cabinet, spray reagents, *(Calypsoarett's)* Calpurnia aurea leaves, 0.1N hydrochloric acid, chloroform, methanol, ethyl acetate, toluene, hair drier, ammonia solutions, Dragendorff's reagent, Iodine tank 1% vanillin solution in sulfuric acid, .

Procedure:

1. Preparation of the chromatoplates

1. Clean five 20x20 cm plates .
2. Place these on the spreader contiguously. Make sure that there are end-plates on both ends and fasten the plates to the spreader.
3. Prepare slurry of the absorbent with water in an approximate ratio of 1:2 by taking about 5 g of silica gel per plate.
4. Place the applicator on one of the end-plates in such a way that it gives a thickness of 0.25 mm in the direction it is to be dragged.
5. Pour the slurry in to the applicator and drag to the end-plate on the opposite end. Allow the slurry to settle on the plate for about 5 minutes.
6. Transfer the plates carefully to a TLC rack and dry the plates in an oven at 100-110°C for one hour.

7. Remove the plates from the oven, cool and if not used immediately, place in a desiccator.

II. Preparation of the sample solution

1. Take the dried extracts from the previous experiments.
2. Dissolve them in a minimum quantity of the respective solvents they were extracted with.

III. Preparation of solvent systems

1. Prepare 200 ml of the solvent system consisting of Chloroform, Methanol and strong ammonium hydroxide solution in the ratio of 90:10:1 respectively
2. Pour the prepared solvent system in to a clean and dry TLC tank with an air tight lid
3. Line the wall of the tank with chromatographic paper and allow it to saturate.

IV. Spotting the sample solution and developing the chromatogram

1. Make a thin line with a pencil 20 mm from one end of the prepared TLC plate. Take care not to scratch or damage the coated silica gel.
2. Apply spots of the prepared sample on the drawn line in such a way that each spot has a diameter no greater than 6 mm. The applied spots should be equidistant.
3. Allow each spot to dry after application. Also, take care not to overload each spot.
4. Place the TLC plate in to the tank and develop the chromatogram until the solvent travels two-third of the height of the plate.
5. Remove the plate and immediately mark the solvent front. The plate is then dried.

IV. Detection of components resolved

1. Observe the plate to see if any colored spots have been resolved and circle them.
2. Observe the plate under UV light and circle any spots observed
3. Spray the plate with Dragendorff's reagent, dry and circle any spots made visible. (x)
4. Record all your observations and calculate the R_f value of all spots observed.
5. Trace your chromatogram and submit it along with your lab report.

EXPERIMENT IV

* Starch Isolation & Chemical Tests

Introduction: Most of higher plants contain starch. The amount of this polysaccharide found in any tissue at any specific time is determined by the physiological activity of that tissue. The starch content of various plants and of the different tissues of the same plant may vary to a great extent. Starch accumulates and becomes especially abundant in seeds, bulbs and tubers. The structures of these organs vary and they may contain substances interfering with starch isolation.

Starch granules are produced within living particles, called plastids, which are found embedded in the cytoplasm of the plant cell. The extraction of starch from plant tissues thus involves the killing of plant tissues. As a result of this, the vacuolar sap will mix with the cytoplasm which in turn affects the quality of starch extracted. Also part of the protein-containing ground substances of the plastids is enclosed by the starch substance as it crystallizes, and other cell proteins adhere to the isolated granule in the form of shreds thus also affecting the quality of starch to be extracted.

Starches are used as components and/or processing aids in the manufacture of products such as food, textile, paper, adhesive and pharmaceuticals.

A number of methods of starch determination are reported in the literature. Below procedures for the extraction of potato starch and wheat starch in addition to the general method for the isolation of starch have been outlined. Ideally, starch extracted using these procedures should be characterized to establish its quality. This characterization includes determination of ash, phosphorous, protein, sulfur dioxide, sulfur, moisture content, fatty substances, alkali number and reducing value. In addition to the extraction procedure, identification and chemical tests for starch are also outlined.

Apparatus and Materials: Potato tubers, wheat flour, teff, distilled water, razor blade, mortar & pestle, centrifuge, muslin cloth, beaker, porcelain dish, oven, water bath, iodine solution, starch, test tubes, Fehling's solution, hydrochloric acid, sodium hydroxide, mercuric chloride solution, ammonium oxalate solution, pH paper.

Procedure:---

I. Isolation of starch from potato

1. Weigh about 200 g of cleaned & washed potato tubers.
2. Reduce to fine pulp using a razor blade and by grinding with mortar & pestle
3. Add enough water to facilitate disintegration and strain through muslin cloth while agitating the pulp continuously with spatula.
4. Express water from the pulp as completely as possible
5. Add more water to residue on the cloth. Repeat the straining, agitating and washing as before
6. Combine all the filtrates, allow it to settle for an hour and decant the supernatant,

7. Wash the residue with water, allow it to settle and decant the supernatant
8. Transfer the residue to a previously weighed porcelain dish and dry in an oven at 50°C .
9. Calculate the percentage yield of starch obtained.

II. Starch recovery from wheat flour

1. Prepare a stiff dough by mixing thoroughly 100 g of flour with 70-80 ml of distilled water
2. Age the dough ball in air at room temperature for 1 hr while covering it with moist muslin cloth.
3. Wash the dough ball by kneading it with hand under a stream of distilled water over nylon cloth drawn over the mouth of a large beaker
4. The slurry collected in the beaker is wet-sieved twice through the bolting cloth
5. Centrifuge the slurry at relative centrifugal forces between 1,500 and 2,300 x G for 10-15 mm.
6. The upper, pigmented fraction, designated as tailings, is carefully removed by scraping with a metal spatula.
7. Purify the residue further by resuspending it in distilled water, centrifuging and again scraping off any tailings remaining. Usually 3 such cycles will result in a good quality starch.

III. General Method for Starch Isolation

1. Remove all non-starchy tissues from the fresh plant material
2. Homogenize the tissue in a mortar (or a blender such as a waring Blender) with the possible addition of 0.01M mercuric chloride solution or cold 1% ammonium oxalate solution
3. Filter the suspension through muslin and centrifuge the homogenate immediately at 200-300xG.
4. Remove crude impurities from the starch layer by repeated gentle washing, followed by resuspension of the starch in water and centrifugation at 1,500xG.
5. If impurities settle down below the starch layer, it is recommended that the starch be resuspended in a small amount of water and centrifuged for a few seconds.
6. Repeat step 5 several times, and each time pour the suspension into a clean tube.

Tests on Starch

A. Identification Test of Starch

1. Boil a very small quantity of the starch you obtained with 15 times its volume of the water
2. Continue boiling until a translucent viscous fluid or jelly is obtained
3. Add iodine solution dropwise and note any change
4. Go on boiling for some time and note any change.

2.0 g
0.05% I₂ soln

\$60

B. Hydrolysis of Starch

1. Weigh 0.6 g of the starch provided and place it in a 125 ml Erdemeyer flask
2. Prepare a paste with a few milliliters of water and then add 40 ml of boiling water.
3. Boil the solution gently until all starch particles disappear and allow the opalescent solution to cool.
4. Divide the solution into two for enzymatic & acid hydrolysis experiments.

i. Enzymatic hydrolysis

1. Place the first half of the starch solution in a test tube
2. In six separate test tubes add 2 drops of 0.002M iodine solution. Label them 1-6.
3. To the first tube add 0.5 ml of starch solution and note the color. This is the zero time sample.
4. Noting the time, add about 0.2 ml of saliva to the starch solution in the test tube and mix the solution well.
5. In 30-second intervals remove 0.5 ml samples from the starch solution and add them to successive tubes of iodine solution.
6. Compare the color of successive samples by observing against a white background
7. Note the time when the iodine solution becomes lavender brown in color
8. Remove 1 ml of starch solution with a pipette transfer it to dry test tube and heat the sample to boiling
9. Cool the test tube under tap water and add 1 ml of fehling's solution. Note the color
10. Again heat the test tube in a boiling water bath for 2 minutes and note the color
11. Remove further samples after 15, 30 and 60 minutes, boil each as it is removed and label with the time.
12. Repeat steps 9 and 10 on the three test tubes in step 11. Compare the colors obtained in the four test tubes.

ii. Acid hydrolysis

1. Add 0.5 ml of conc. hydrochloric acid to the other half of the starch solution prepared
2. Arrange a series of empty test tubes and then place the starch solution on a boiling water bath. Note the time.
3. At 5-minute intervals withdraw 0.5 ml of sample from the starch solution being boiled and put it in an empty test tube
4. Cool to room temperature and add 2 drops of iodine solution
5. Remove further 1 ml samples after 15 and 30 min. heating and also 10 minutes after the iodine starch test gives a light brown color.
6. Add a few drops of 2N sodium hydroxide to the above three samples (until neutral to indicator).
7. To each of the three samples add 1 ml of Fehling's solution
8. Heat the three test tubes for 2 minutes and compare the colors produced.

EXPERIMENT V

Qualitative Tests for Carbohydrates

Introduction: Carbohydrates are, in general, aldehyde or ketone, alcohols or are condensation polymers of these partially oxidized polyalcohols. They can be conveniently classified into two main groups according to their complexity.

- i. The sugars, which are carbohydrates that contain one sugar unit (saccharide) that cannot be hydrolysed or are condensation products of about 2-10 of these sugar units.
- ii. The polysaccharides, which are a heterogeneous group that yield a large and indefinite number of monosaccharide units on hydrolysis.

Carbohydrates consist of at least one unit of a monosaccharide which is a polyhydroxy compound containing either a potential aldehyde or ketone group. They thus may exhibit, at least the simple sugar unit, the usual reactions of alcohols and of aldehydes or ketones. In addition, they undergo many reactions that are not given by simple alcohols or carbonyl compounds. Basically most of the qualitative tests outlined below make use of these functional groups.

Apparatus and Materials: glucose, fructose, galactose, arabinose, ribose, lactose, sucrose, maltose, starch, cellulose, Fehling's solution A & B, Benedict's solution, Barfoed's solution, α -naphthol, sulphuric acid, anthrone reagent, Bial's reagent, Seliwanoff's reagent, iodine-potassium iodide solution, amyl alcohol, test tubes, water bath.

Procedure:

Prepare a series of solutions of the following compounds and subject to each test. Also, prepare two concentrations, 0.1% and 1% of each solution, so that the sensitivity of the test can also be assessed: glucose, fructose, galactose, arabinose, ribose, lactose, sucrose, maltose, starch and cellulose.

I. Fehling's Test:

1. weigh 7 g of $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$ in water and make up to 100 ml
2. Weigh 24 g of KOH and 34.6 g of sodium potassium tartrate and place in a separate container, dissolve in water and make up to 100 ml.
3. Mix the two solutions just prior to use
4. To 2 ml of this solution add a few drops of the test solution and boil in a water bath. Note any changes.

*3 - CuSO_4
24 - KOH, tartaric acid, NaOH*

II. Benedict's Test

1. Weigh 17.3 g of sodium citrate and 10 g of sodium carbonate and dissolve in about 75 ml of water (filter if necessary).
2. Separately weigh 1.73 g of $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$ and dissolve in about 20 ml of water
3. Add the solution prepared in step two slowly with stirring to the alkaline citrate solution i.e the solution prepared in step one. Make up the volume to 100 ml
4. To 2 ml of this solution add a few drops of the test solution and boil in a water bath. Note any changes.

*CuSO_4
Na₂CO₃
Na citrate*

III. Barfoed's Test

1. Weight 13.3 g of copper acetate and dissolve in 200 ml of water.

- reduce monosaccharide // reduce disaccharide

CuSO_4 in acid

→ red for mono

*copper acetate
+ acetic acid*

To the above solution add 1.8 ml of glacial acetic acid
 To 2 ml of this reagent add 1 ml of test solution and boil exactly for 1 minute and note any changes.

Molish's Test
 Take 2 ml of test solution in a test tube and add 2 drops of an ethanolic solution (50%) of α -naphthol.
 Incline the tube at 45° angle and carefully let 1 ml of conc. H_2SO_4 run down the sides of the tube without mixing. Note any change at the junction of the two liquids. If a suspension was used, shake after the addition of acid.

Anthrone Test
 Dissolve 200 mg of the anthrone reagent in 100 ml of conc. H_2SO_4
 Take 2 ml of this solution in a test tube, add 2 drops of the test solution and mix well
 If there is no immediate color change, boil in a water bath for 10 minutes

Bial's Test
 Weigh 150 mg of orcinol and dissolve in 50 conc. HCl
 Mix 5 ml of this reagent and 2 ml of the test solution in a test tube and heat in a water bath.
 Note the time at which any color change is observed: if it happens before 10 minutes it is a pentose.

Seliwanoff's Test
 Weigh 50 mg of resorcinol and dissolve in 100 ml dilute HCl (3N or 1:2 diluted)
 To 5 ml of this reagent add a few drops of the test solutions and heat the mixture to boiling and observe and record any change.
 Add few drops of amyl alcohol, observe and record if any change occurs.

Iodine Test
 Prepare 0.005N iodine solution in 3% potassium iodide
 Take a few drops of the test solutions in a test tube, add 1 or 2 drops of 0.1N HCl followed by 2 drops of the solution prepared in step one.

Scheme for Analysis of an unknown Carbohydrate

Test	Observation	Conclusion
Physical Character		
Condition		
Colour		
Odour		
Taste		
Solubility (H_2O) alc... etc)		
Chemical properties		
Action of dry heat (on the solid)		
Molish's test (on the solid)		
Action of KOH solutions & heat		
Action I_2 solution		
Test for oxidase enzyme (Benzidine + H_2O_2) (on the solid)		
Reduction tests Fehling's reagent A & B		
Bartoe's Test		
Test for pentozans (on the solid and after acid hydrolysis)		
a) Aniline acetate test ✓		
b) Action & phloroglucin ✓		
c) Action of acetone /HCl		

9	Test for methyl pentanes e.g rhamnose a) Aniline acetate paper b) Action of active /Hcl		
10	Tests for ketohexoses a) Cobalt nitrate test b) Seliwanoff's test (resorcinol + Hcl+ heat) c) Bromine water test(1-2 drops)		
11	Tests for aldohexoses a) Bromine water (1-2 drops) b) $\text{Ca}(\text{OH})_2$ test solution c) Osazone formation Reaction with (hydroxylamine Hcl or phenyl hydrazine)		
12	Test for Polysaccharides Acid-hydrolysis: (Hcl & heat)		
13	a) Fehling's test (A & B) b) I_2 Solution test c) Test for ketoses d) Test in pentozans		
	Enzymatic hydrolysis Saliva+ H_2O and heat a) Fehling's test (A & B) b) I_2 Solution test c) Test for ketoses d) Test in pentozans		
	Tests for Gums a) Oxidase enzyme test b) Action of lead acetate c) Action lead sub acetate d) Action of Borax solution		
The unknown carbohydrate is :			

EXPERIMENT VI

ANTHRAQUINONE GLYCOSIDES

Introduction

These glycosides, which possess anthracene or its derivatives as aglycone, are called Anthraquinone glycosides or Anthraglycosides. The principal examples of crude drugs containing anthraquinone glycosides are cascara, rhubarb, aloe and senna. Anthracene compounds can occur in these plant materials as free aglycones or as glycosides, which can either be O-glycosides or C-glycosides. Simple acid hydrolysis is possible for O-glycosides while the glycosidal linkage needs oxidative hydrolysis.

Apparatus and Materials: plant material test tubes, dilute ammonia solution, benzene, FeCl_3 T.S., NH_4Cl , borax solution, aloe powder, Kieslgulur, bromine water.

Procedure:

Detection of free anthraquinone glycosides in a plant material

Borntrager's test

Weigh about 5g of the powdered plant material and triturate it with about 25ml of benzene for about 10 minutes and filter. Keep the marc for the next experiment. Take about 5ml of the filter in a clean test tube and carefully add 3ml dilute ammonia solution. Shake the mixture vigorously and note the color of the aqueous phase when the two separate.

Detection of anthraquinone glycosides in a plant material

Modified Borntrager's test for O-glycosides

Transfer the marc obtained in (I) into a 250ml conical flask and add 50ml of 2N HCl and reflux for 30 minutes on a hot plate. Cool the flask under running water and filter. Extract the filtrate with three 25ml portions of benzene. Check if all the O- glycosides are extracted out by extracting the filtrate with a further 15 ml of benzene and by adding few drops of dil. Ammonia to the extract. If they are still present, extract the filtrate again with another 25 ml of benzene. Combine all the benzene extracts. Keep the aqueous layer for the next experiment.

Take about 5ml of the benzene extract in a clean test tube and carefully add 3 ml dilute ammonia solution. Shake the mixture vigorously and note the color of the aqueous phase when the two separate.

II. Modified Borntrager's test for C-glycosides

To the aqueous layer recovered from (II) add 3ml of FeCl_3 T.S. and heat the mixture for 15 minutes on a hot plate. Extract the liberated anthraquinones by using benzene. To 5ml of the benzene extract add 3ml dilute ammonia solution. Shake the mixture vigorously and note the color of the aqueous phase when the two separate.

General Chemical Tests for Aloes - DL

For the following tests boil 1g of the drug in 100ml of water, add a little Kieslguhr and filter. Use separate portions of the filtrate for the following tests.

1. Schottgen's Reactions: To 5ml of solution of aloes add 2ml of borax solution and heat on water bath for 15 minutes. Pour a few drops of the liquid into a test tube nearly full of water. Observe the mixture under UV light and record the fluorescence produced.

2. Bromine Test: To two ml of solution of aloes add 2ml of freshly prepared solution of bromine and observe for production of yellow colored precipitate.

Specific Chemical Tests

* Nitric acid test: To 5ml solution of aloes add 2 ml of nitric acid. Cape aloes gives a _____ color rapidly changing to _____, and Curacaoa _____ color.

4539/00
mehmet Z

EXPERIMENT VII

Saponins

The group of glycosides known as saponins is, in varying degrees, characterized by two common properties, i.e. the saponins foam in aqueous solution and they cause haemolysis of red blood cells. The aglycons of saponins are known as Sapogenins and according to the chemical nature of the sapogenin, a saponin may be classified as a steroidal saponin or a triterpenoid saponin. This group of complex glycosides is of common occurrence, being found in more than fifty plant families and 400 different species scattered all over the world.

In this experiment, saponins are extracted from an indigenous plant, *Phytolacca dodecandra* (ENDOD), and their characteristics are studied.

Apparatus and Materials: Plant material, balance, measuring cylinder, physiological salt solution containing 2.5% defibrinated blood.

Procedure:

I. Extraction

1. Take 5 g of the powdered plant material-*Phytolacca dodecandra*
2. Add 100 ml of distilled water and boil in a beaker for 4 minutes
3. Filter hot and allow to cool.

I. Qualitative determination:

A. Form Index

1. Place 2 ml of the solution obtained in the extraction procedure (step 3) in a series of test tubes.

2. Prepare different dilutions of the aqueous extract placed in the test tubes.
3. After dilution, shake each test tube vigorously for 15 seconds.
4. Allow to stand for 15 minutes and measure the height of the foam layer. The dilution which gives a 1 cm high foam after 15 minutes signifies the foam index.

B. Hemolytic Method (Hemolytic Index)

1. Place 5 ml of the solution obtained in the extraction procedure (step 3) in a series of test tubes.
2. Prepare varying dilutions of the extract placed in the test tubes above.
3. Prepare a physiological salt solution containing 2-5% defibrinated blood.
4. Place 4 ml in a series of test tubes.
5. 2 ml of the varying dilution prepared in step-2 are added to the physiological salt solutions (containing defibrinated blood) arranged in step 4.
6. Note any changes. Haemolysis takes place when sufficient saponin is added, i.e; the blood suspension then becomes transparent. The weakest dilution of saponin causing total haemolysis is the so-called hemolytic index.

EXPERIMENT VIII

LIPIDS

Lipids form a heterogeneous group of compounds which have the common property of not dissolving in water but dissolving in organic solvents such as alcohol, chloroform, benzene, ether and hexane. Other than this hydrophobicity no other common features can be attributed to their structures.

Some of the most common lipids are fixed oils and fats. Chemically they are esters of the tri-hydric alcohol glycerol with long chain fatty acids. The isolation procedure depends particularly upon how much oil is present, on the degree of purity required in the product and on the nature of the raw material. Extraction of pharmaceutical oil is normally carried out by expression. However, solvent extraction is also sometimes used to obtain cheap technical grade oil. In this experiment organic solvents are used for the extraction of oils from local oil bearing seeds.

Material

Sesame seeds (*Seasma indicum*), Pea-nut (*Arachis hypogaea*), castor beans (*Ricinus communis*) and nug seeds (*Guizolia abyssinica*).

Procedure

1. Weigh 100g of the plant material provided and extract the oil according to procedures given below:

a. For sesame and nug seeds

- (i) Grind the seeds in a mortar and reflux with benzene for one hour
- (ii) Add activated charcoal to remove coloring matters and filter
- (iii) Remove the organic solvent with a rotavapour
- (iv) Cool and add about 50 ml of distilled water
- (v) Transfer the mixture to a separatory funnel and separate the oil
- (vi) Calculate the percentage yield as V/W

b. For castor beans and pea-nuts

- (i) Remove the husks and crush the seeds into smaller pieces
- (ii) Extract the oil with benzene using soxhlet for one hour
- (iii) Add activated charcoal to remove coloring material and filter
- (iv) Remove the organic solvent with a rotavapour and calculate the percentage yield as V/W

WRITING A SCIENTIFIC REPORT

The format to write a scientific report has the following general structure.

1. **Date, Experiment No. and Title:-** should be specified at the top.
2. **Objective:-** the objective of the experiment should be stated in a perfectly clear way.
3. **Background:-** a summary of the theoretical background of the practical aspect should be written.
4. **Method:- materials & Procedures**
The method is a carefully designed procedure which yields appropriate data allowing to draw inferences from them. To fulfill the primary conditions, the method should describe the materials and the procedure adopted in complete detail (names and grades of the materials used, concentration, pH, temperature and the method of assay should be mentioned).
5. **Results:-** refer to the data obtained from the experiment and may be represented by a table or a graph.
 - a) **Tabular representation:-** the corresponding values of an independent variable (x) and a dependable variable (y) should be tabulated. The table should be clearly titled and each column should have a heading giving the names and units of the quantity listed.
 - b) **Graphical representation:-** the first step in preparing a satisfactory graph is to select the correct graph paper.
 - Rectangular coordinate paper is satisfactory for most purposes.
 - Semilogarithmic paper is convenient when one of the coordinates is to be the logarithm of an observed variable.
 - Log-log paper may be used, if both coordinates are log of variables.The names and the units of each quantity should be given along each axis. Each plotted quantity should be given along each axis. Each plotted point should be surrounded by a circle or a suitable symbol. After the points are plotted, the best smooth curve is drawn through the points. If it is not possible to draw the line with any degree of confidence, a better means is to carry out a *regression analysis*.
- Discussion:-** the result obtained or inferred should be evaluated and matched against the objective proposed and questions should be answered.
- Conclusion:-** is made based on the results obtained and in relation to the objective.

GRAPHS AND REGRESSION ANALYSIS

A graph is one of the most illuminating ways of presenting results. In investigations of the influence of one variable on another (e.g. effect of pH on the reaction rate constant k of a chemical decomposition reaction), the magnitude of the independent variable (pH in the above example) is customarily indicated on the horizontal axis or abscissa and the vertical axis or ordinate is used for the dependent variable (k in the above example). Selection of appropriate scales for the coordinates should be based on a reasonable compromise between convenience and clarity. Once all individual points have been plotted, a line is drawn to join the points in sequence. For data described by a linear equation of the form: $y = A + Bx$, the straight line that fits the data better than any other line can be obtained by *regression analysis* and is known as the *regression line*.

Regression analysis is a statistical method used to analyze experimental results showing linear relationship of one variable to another, e.g., the extent of hydrolysis of an ester drug as a dependent variable and time as an independent variable. The first step in the analysis is to determine whether or not the experimental data should fit a straight line. For this, the correlation coefficient (r) is calculated using the equation:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2}}$$

A value of zero indicates no correlation at all, while the limiting values, +1 and -1, indicate either perfect positive correlation or perfect negative correlation. Assuming that the calculated value of r shows a significant correlation between the two variables, the experimental points can be related by a straight line. The most probable locus of this line can be determined by calculating the slope (B) or regression coefficient and intercept (A) of the line using the following equations:

$$B = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} \quad \text{and,} \quad A = \bar{y} - B\bar{x}$$

The fitted regression equation is thus: $y = A + Bx$

In conclusion:

- 1- Linear regression can be properly applied only after it has been shown that the two variables are correlated and that the relationship between them is linear.
- 2- The method is used to fit the most probable straight line to experimental data showing relationship of two variables.
- 3- In addition, the method allows to predict the magnitude of one of the variables at a predetermined value of the other.
- 4- Regression analysis is not used to increase the precision of the experimental data and hence does not reduce the scatter of experimental points. Moreover, the analysis is independent of the type of variables considered in various experiments.